# PETROS

VERSION 4.0.0

Comprehensive End-User Manual & Operational Handbook

## 1. System Overview

Petros is an integrated AI-powered engineering environment that unites local intelligent reasoning, autonomous multi-agent software development, Web3 cryptographic wallet security, and rapid application generation in a unified, privacy-first desktop application.

## 2. The Petros Ecosystem Rooms

| Room / Module | Purpose & Core Capabilities |
|---|---|
| **Laya AI Intelligence** | Persian & English conversational co-founder, strategic startup advice, requirement refinement, architecture planning. |
| **Studio (Engineering)** | Multi-file code editor, real-time code synthesis, terminal execution, and Git revision history. |
| **Laboratory** | Interactive testing suite, automated benchmark verification, syntax validation, and test harness execution. |
| **Agency (Agent Squad)** | Autonomous engineering team orchestration: Supervisor, Coder, Writer, Router, and Human-in-the-loop Gatekeeper. |
| **Commerce & Fuel** | Project credit tracking, compute fuel metrics, and transparent usage allocation. |
| **App Factory** | Rapid visual React application generator with in-memory TSX transpilation and one-click workspace export. |

## 3. Conversational Intelligence with Laya

Laya is Petros's built-in engineering and startup AI co-founder. Laya automatically adapts to your preferred language:

- **Dual-Language Fluency:** Seamlessly converses in native Persian (فارسی) or English.
- **Local-First Inference:** Routes requests through local Ollama models (e.g. `qwen3:1.7b`) on port 11434 with **$0.00 API expenditure**.
- **Context Retention:** Retains conversation history and architectural decisions across sessions via the Mem0 project memory store.

## 4. Web3 Wallet Authentication & Security

Petros uses decentralized cryptographic identity rather than traditional passwords:

1. **Generate or Connect:** Create a new 12-word mnemonic BIP-39 wallet or connect an existing Ethereum address.
2. **Zero Plaintext Storage:** Private keys and recovery phrases are never stored in plaintext or transmitted across networks.
3. **Cryptographic Challenge:** Session tokens are authenticated via EIP-191 ECDSA signatures with single-use nonce validation and replay protection.
4. **Vault Encryption:** Local secrets and API tokens are sealed using AES-256-GCM authenticated encryption.

## 5. Autonomous Agent Squad Orchestration

When developing complex software projects, Petros deploys specialized autonomous agents:

- **Supervisor:** Analyzes requirements, breaks tasks into structured phases, and oversees project execution.
- **Coder:** Generates production TypeScript, React, and Node.js source code.
- **Writer:** Authors technical documentation, OpenAPI specifications, and release notes.
- **Router:** Evaluates and cascades model routing according to performance and privacy constraints.
- **Human-in-the-Loop Gate:** Solicits human approval before committing code changes or running system commands.

## 6. App Factory & Project Export

App Factory empowers you to generate full-stack web applications from natural language prompts:

1. Describe your desired application in the App Factory prompt input.
2. App Factory generates reactive TSX components with instant in-memory transpilation and live interactive preview.
3. Click **Export Project** to export a clean, standalone, npm-ready project directory directly into your workspace at `%USERPROFILE%\Desktop\petros\apps\generated\`.

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